

Motion-Adaptive Internal Force Allocation for Dual-Arm Cooperative Manipulation // Or // Task-on-Demand Grasp Force Optimization for Dual-Arm Object Transport

1st Luca Morello
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

2nd Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

3rd Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

Abstract—ToDo
Index Terms—ToDo [1]

I. INTRODUCTION

- Dual-arm cooperative manipulation needs enough squeeze force to prevent slip, but excess squeeze force is wasted effort / risks damaging or destabilizing fragile objects.
- Human grip-force control scales force with load/motion, not statically (Westling & Johansson 1984)
- Most cooperative-manipulation force-distribution work optimizes internal force *statically or per-instant*; most motion-adaptive grip-force work is single-gripper/slip-triggered, not posed as a dual-arm allocation problem.
- **Gap:** no existing framework couples
 - 1) nullspace-based internal/external decomposition,
 - 2) real-time QP allocation between two arms
 - 3) motion-derived force demand that ramps up specifically during object transport and relaxes at rest.
- Contributions
 - 1) A QP formulation for real-time squeeze-force allocation between two cooperating arms with friction-aware bounds.
 - 2) A motion-adaptive demand term that couples the desired internal force level to the object's trajectory state.
 - 3) Hardware/sim validation on a dual xArm7 platform (mc_rtc).
- D. (brief) Impact-aware contact establishment — cited as context for your grasp-initiation phase, explicitly *not* claimed as a contribution here.

TABLE I
TABLE TYPE STYLES

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ACKNOWLEDGMENT

REFERENCES

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- [1] M. Uchiyama and P. Dauchez, "A symmetric hybrid position/force control scheme for the coordination of two robots," in *Proceedings. 1988 IEEE international conference on robotics and automation.* IEEE, 1988, pp. 350–356.

II. RELATED WORK

- A. Internal/external force decomposition and QP-based load distribution in dual-arm systems.
- B. Friction-cone-constrained grasp/contact force optimization (exact SOCP vs. linear/QP approximations).
- C. Motion- and slip-adaptive grip force control (bio-inspired and robotic).